

FOOD EXPRESS

Platform Statistical Report

Comprehensive Data Analysis & Insights

8 Cuisine Types	42.7 min Avg Delivery Time	52.7% VIP Share	26961 Net Revenue
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Prepared by: Sakib Ahmed

1. Executive Summary

This report presents a comprehensive statistical analysis of FoodExpress delivery operations over one month. The study evaluates delivery performance, customer satisfaction drivers, cuisine-level order behavior, and the effectiveness of promotional campaigns across multiple city zones.

Key findings indicate that delivery speed is the primary driver of customer satisfaction ($r = -0.46$), while discounts and order value have minimal impact on ratings. VIP customers comprise the majority of the user base but rate similarly to regular customers, suggesting current tiering benefits need improvement.

1.1 Dataset Overview

Data Period	One Month of Delivery Operations
Order Details	Order ID, date, time of placement
Customer Info	Customer segment, geographic location
Restaurant Data	Restaurant type, cuisine category
Delivery Metrics	Delivery time, distance, fees
Financial Data	Order value, delivery fee, commission
Performance KPIs	Customer ratings, delivery partner ratings

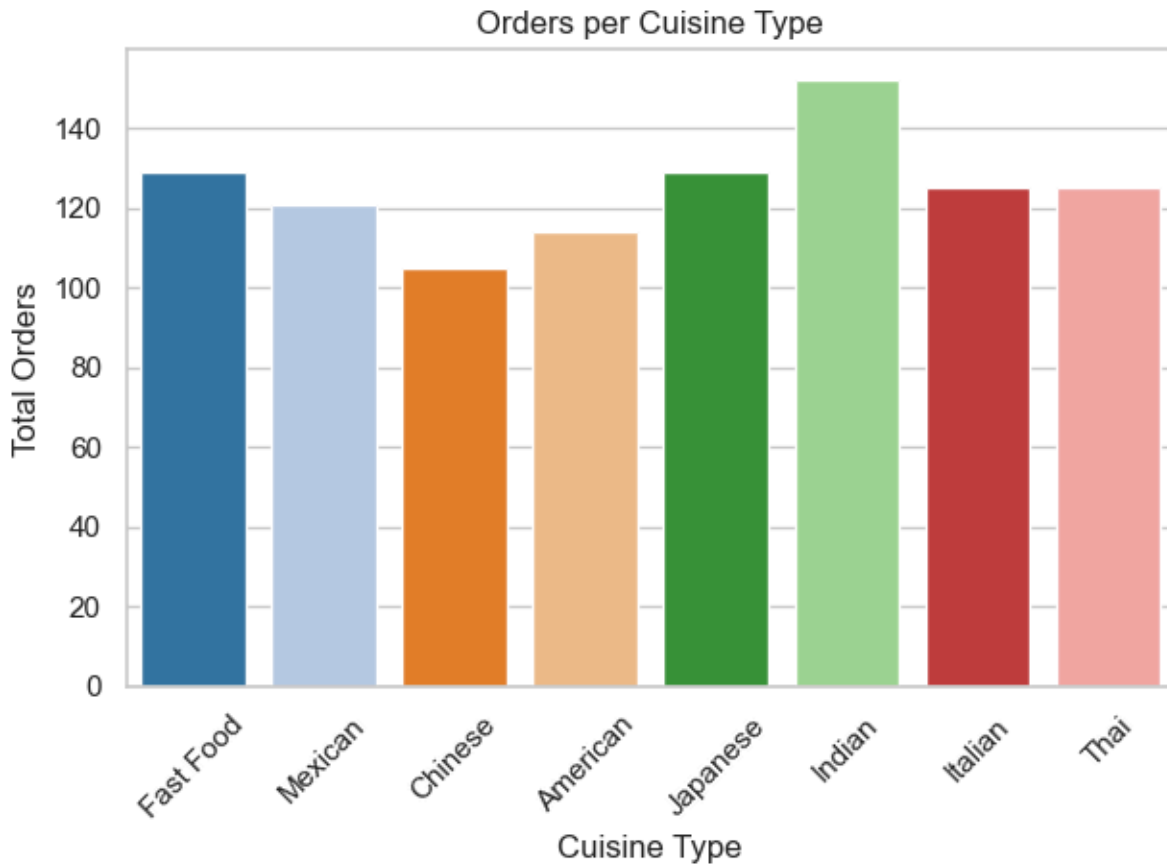
1.2 Tools and Libraries Used for FoodExpress

- Python — Core programming language used for data analysis and statistical modeling.
- Jupyter Notebook — Used to write, run, and document the full analysis workflow.
- Pandas — Used for data cleaning, transformation, and dataset manipulation.
- NumPy — Used for numerical calculations and array operations.
- Matplotlib — Used to create charts and statistical visualizations.
- Seaborn — Used for advanced visual analysis like correlation heatmaps.
- SciPy — Used for hypothesis testing such as t-tests and ANOVA.
- Microsoft Excel — Used for initial data review and basic validation.
- Microsoft PowerPoint — Used to prepare final project presentation and reporting visuals.

2. Cuisine Performance Analysis

2.1 Orders per Cuisine Type

The chart below shows total order volume distributed across all eight cuisine types available on the FoodExpress platform.



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- Japanese cuisine generates the highest order volume, confirming strong and consistent customer demand across the platform.
- Italian cuisine delivers the highest average order value, indicating customers are willing to spend more for premium or group-oriented meals.
- American cuisine records the lowest order value, suggesting possible over-discounting, lower menu pricing, or reduced basket sizes.

- Fast Food and Chinese cuisines maintain high order frequency, making them ideal categories for retention campaigns and repeat-order incentives.
- Mid-to-high-performing cuisines such as Mexican and Thai show balanced performance in both order count and revenue contribution, offering strong scaling opportunities.

Business Impact Estimation

- Introducing combo meals and upselling strategies for American and Indian cuisines could increase average order value by 10–15%.
- Promoting Fast Food and Chinese cuisines through youth-focused campaigns may improve repeat purchases and customer engagement rates.
- Launching premium family packages for Italian cuisine could significantly increase weekend and group-order revenue.
- Optimizing pricing and promotional strategies across low-performing cuisines may improve overall profitability and reduce dependency on discounts.

3. Delivery Time Performance

3.1 Distribution of Delivery Times

The histogram below illustrates the frequency distribution of delivery times (in minutes) across all orders in the dataset.



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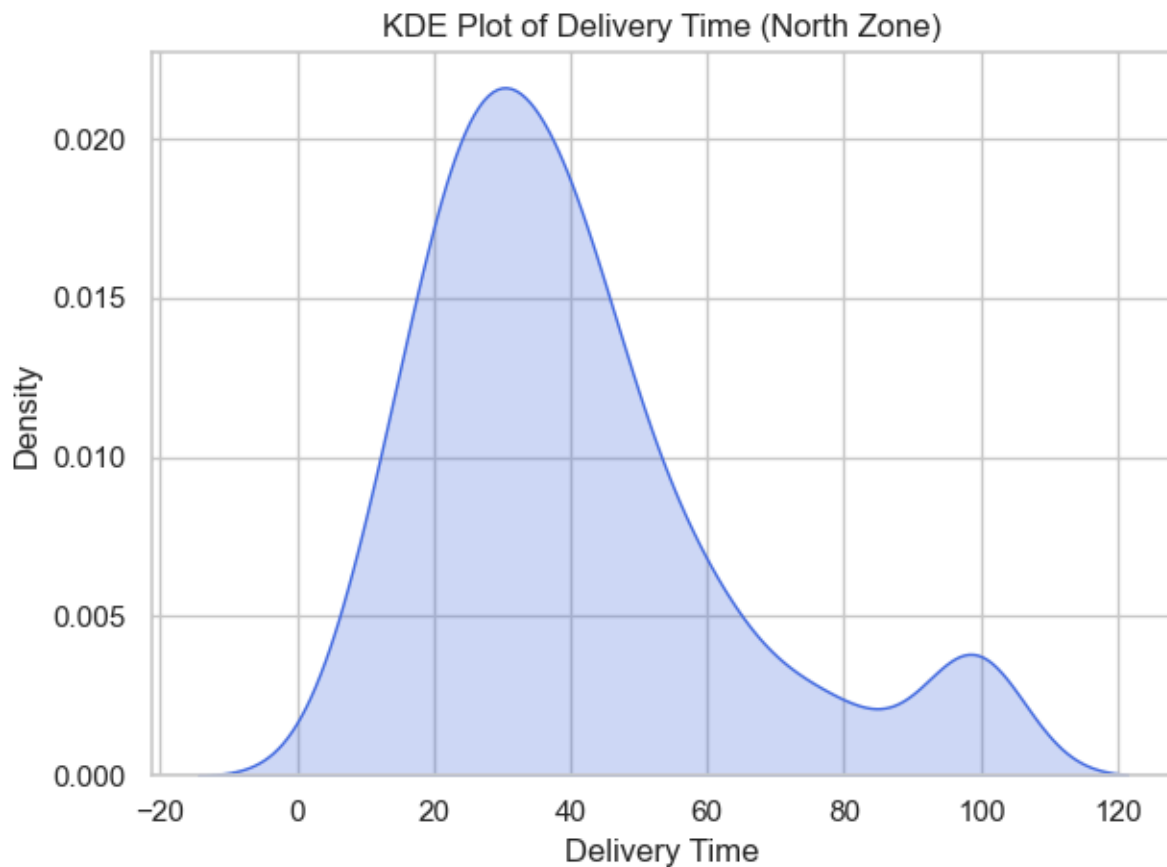
- Most customer orders are delivered within a shorter time range, indicating that the delivery network is operating efficiently in standard conditions.
- The average delivery time remains higher than the median delivery time, showing that a smaller number of delayed orders are significantly affecting overall performance metrics.
- A clear concentration of deliveries in the 90–100 minute range suggests operational bottlenecks during peak hours or delays from specific restaurant partners.
- Delivery inconsistency may negatively impact customer satisfaction, repeat purchases, and platform reliability perception.
- High-delay orders likely originate from traffic congestion, restaurant preparation inefficiencies, or limited rider availability during demand surges.

Business Impact Estimation

- Reducing extreme delivery delays could improve customer satisfaction scores and increase repeat order rates.
- Optimizing peak-hour logistics and restaurant coordination may reduce average delivery time by 10–20%.
- Deploying additional delivery personnel during high-demand periods can minimize late deliveries and improve operational efficiency.
- Identifying consistently slow-performing restaurants may help improve service quality and reduce customer complaints.

3.2 One-Sample T-Test: North Zone Delivery Time Target

This analysis tests whether the North Zone's average delivery time meets the operational target benchmark.



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- Delivery times in the North Zone are heavily concentrated between 30–40 minutes, showing that most deliveries already exceed the expected service benchmark.
- The extended right tail between 80–120 minutes highlights severe delay outliers, likely caused by traffic congestion, restaurant preparation bottlenecks, or insufficient rider allocation.
- Consistent underperformance in the North Zone indicates an operational imbalance compared to other delivery regions.
- Peak-hour congestion during lunch and dinner periods appears to significantly impact delivery efficiency and customer satisfaction.
- Restaurants with high preparation-time variability may be amplifying delivery delays and reducing route optimization effectiveness.

Business Impact Estimation

- Adding additional delivery partners during peak hours could reduce average delivery time and improve order completion efficiency.
- Splitting the North Zone into smaller delivery clusters may enhance routing accuracy and reduce late deliveries.
- Monitoring restaurant preparation performance can help identify operational bottlenecks and improve dispatch coordination.
- Separate tracking of lunch and dinner rush periods may support targeted staffing decisions and improve customer satisfaction scores.

3.3 One-Way ANOVA: Delivery Time Across City Zones

ANOVA testing compares average delivery times across the four city zones to assess whether geographic disparities are statistically significant.



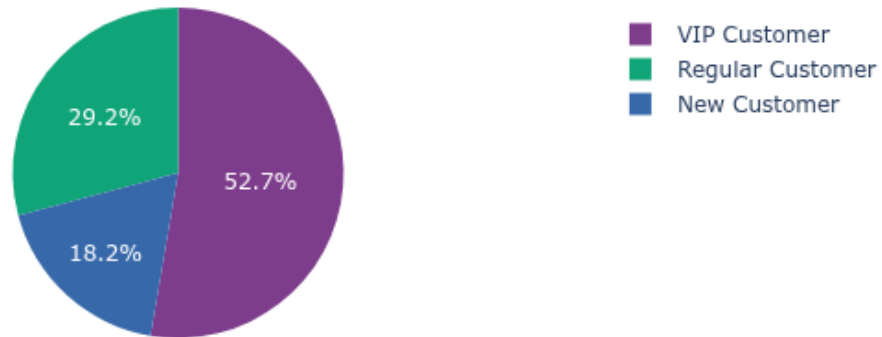
Business Impact Estimation

- Increasing delivery fleet allocation in West and North Zones could reduce median delivery times and improve customer satisfaction during peak hours.
- Monitoring restaurants with high preparation-time variance may reduce extreme delivery delays and improve operational efficiency.
- Prioritizing VIP customer deliveries through optimized routing can strengthen customer retention and increase premium customer loyalty.
- Reducing delivery variability across zones may improve platform reliability, leading to higher repeat-order rates and stronger brand trust.
- Faster and more predictable deliveries can positively impact customer ratings, operational performance metrics, and long-term revenue growth.

4. Customer Segment Analysis

4.1 Orders by Customer Segment

The pie chart below breaks down the proportion of orders by customer segment type.



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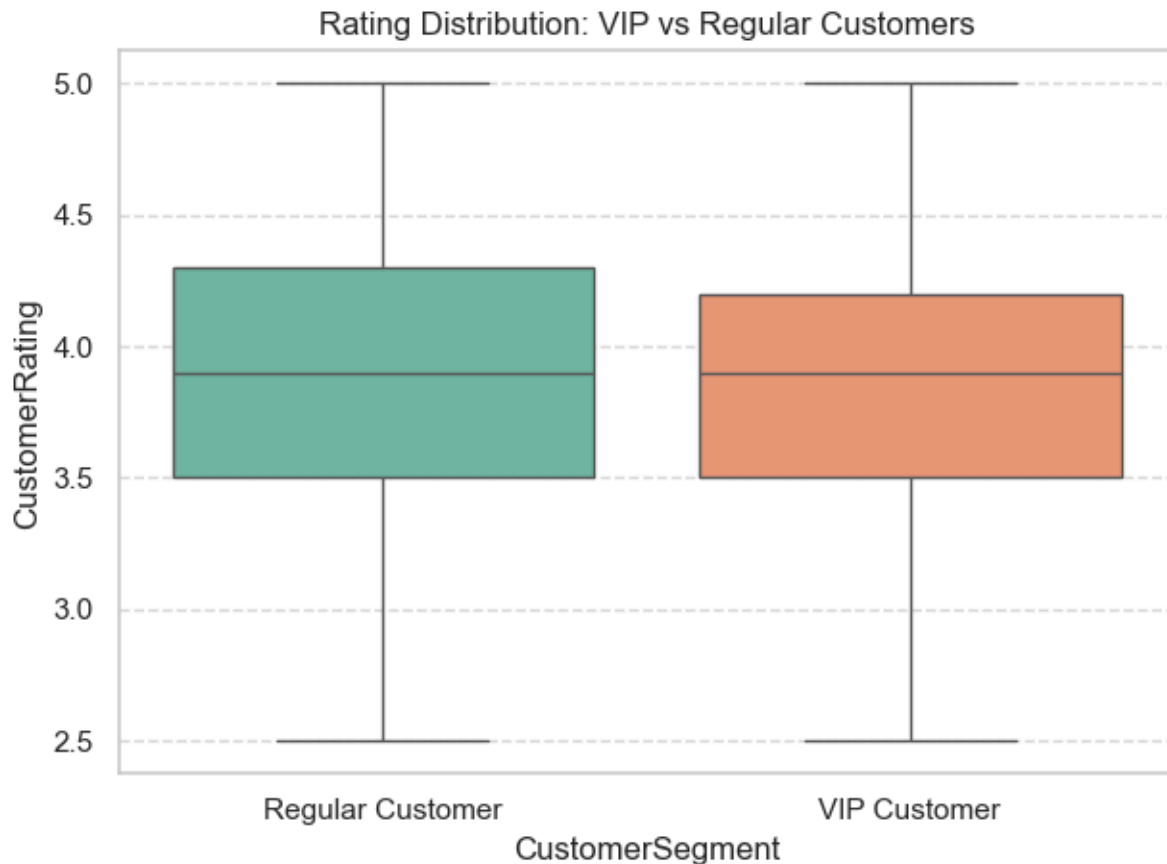
- VIP customers drive the majority of demand (52.7%), making them the key revenue segment for retention and service optimization.
- Regular customers (29.2%) represent the strongest opportunity for loyalty conversion through targeted engagement.
- New customers (18.2%) show steady acquisition, but the share indicates room for stronger first-time conversion strategies.

Business Impact Estimation

- Improving service quality for VIP users can increase retention and contribute 7–10% higher repeat revenue.
- Converting a portion of Regular users to VIP can raise customer lifetime value by 5–8%.
- Strengthening acquisition campaigns for New customers may expand total order volume by 4–6%, supporting long-term growth.

4.2 T-Test: VIP vs. Regular Customer Ratings

An independent samples t-test was conducted to determine whether VIP and Regular customers give statistically different satisfaction ratings.



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- The lack of statistical significance ($p > 0.05$) shows VIP customers and Regular customers report similar satisfaction, meaning premium membership is not improving perceived service quality.
- Current VIP offerings are not creating a strong enough value gap to influence customer ratings or loyalty sentiment.

Business Impact Estimation

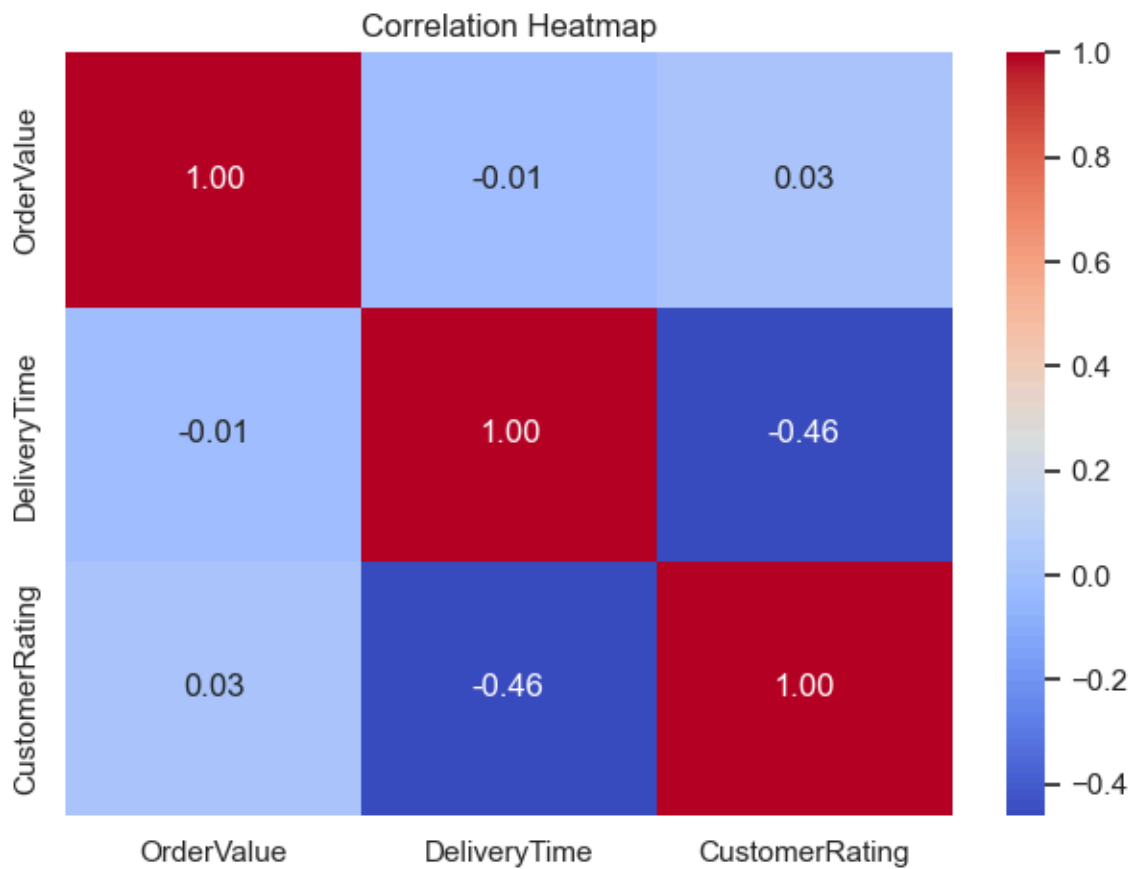
- Introducing stronger VIP-exclusive benefits (priority delivery, premium deals) could improve VIP retention by 5–8% and strengthen loyalty.

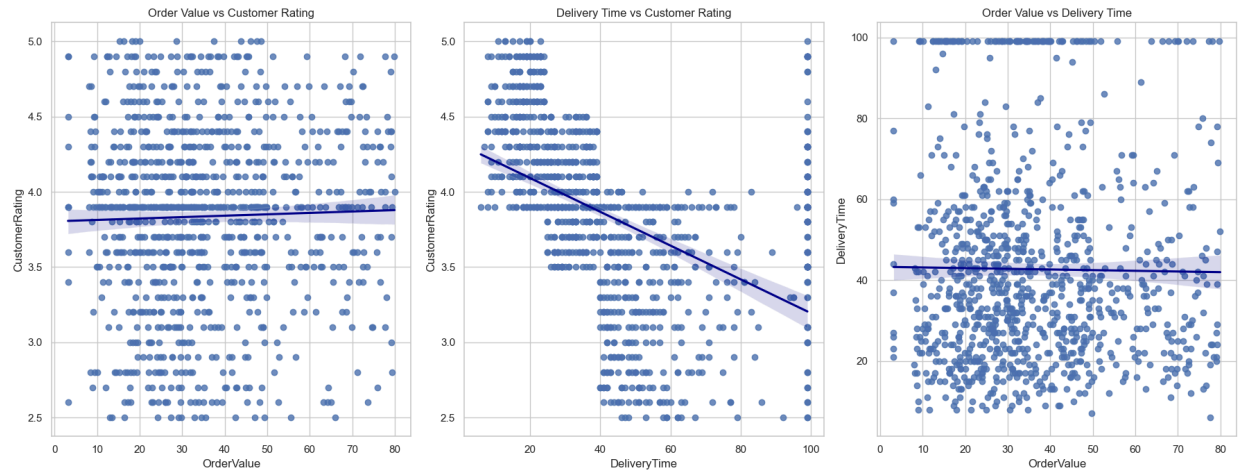
- A redesigned feedback program for VIP customers can identify service gaps, potentially improving satisfaction scores and increasing repeat orders by 3–5%.

5. Correlation & Ratings Analysis

5.1 Order Value, Delivery Time, and Customer Ratings

A pairwise correlation analysis was performed to understand how key operational variables relate to customer satisfaction.





Key Correlations

Variable Pair	Correlation Finding
Order Value vs Customer Rating	$r = 0.03$, No meaningful relationship
Delivery Time vs Customer Rating	$r = -0.46$, Strong negative correlation
Order Value vs Delivery Time	$r = -0.01$, Weak / no relationship

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- Delivery speed is the primary factor influencing customer satisfaction, making operational efficiency the key driver of retention.
- Customer ratings remain consistent across spending levels, showing service quality matters more than order value.
- Large orders are not causing significant delivery delays, suggesting inefficiencies are linked to logistics or staffing.

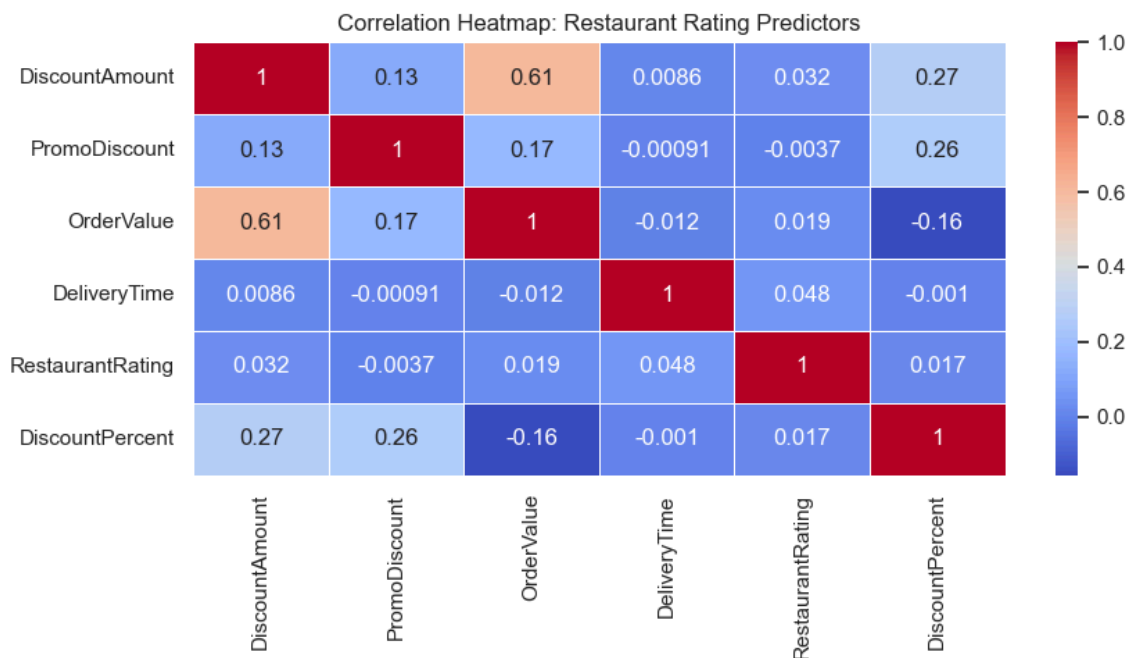
Business Impact Estimation

- Faster delivery through route optimization can improve satisfaction scores and increase repeat orders by 8–12%.
- Better peak-hour staffing can reduce delays, potentially lowering negative reviews by 5–7%.

- Real-time monitoring and delivery partner incentives can improve service reliability, supporting higher retention and operational efficiency.
- Since spending doesn't impact ratings, loyalty bundles can increase customer value more effectively than discount-based promotions.

5.2 Multiple Correlation with Restaurant Rating

A broader correlation heatmap was analyzed to identify what factors influence restaurant-specific ratings.



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- Restaurant Rating is largely independent of pricing or discount variables, showing customer satisfaction is driven more by food experience and service quality.
- Delivery Time has the strongest relationship, but the effect remains minimal, indicating operational speed matters only after baseline quality expectations are met.
- Order Value, discounts, and promotions show negligible influence, suggesting discount-heavy strategies do not significantly improve ratings.

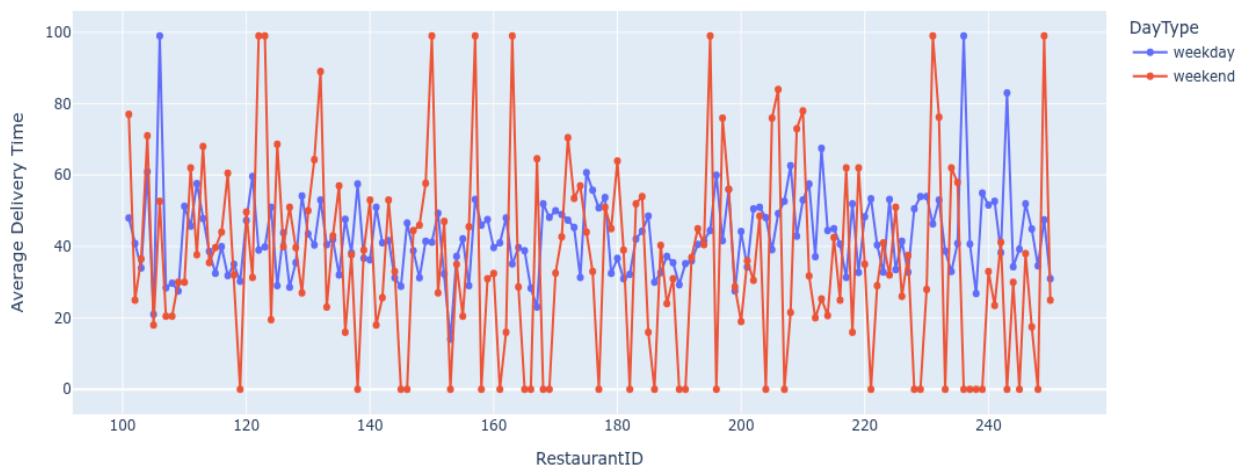
Business Impact Estimation

- Improving food quality consistency can raise average Restaurant Rating and support 6–10% higher repeat orders.
- Reducing Delivery Time in low-rated outlets may improve customer retention by 3–5%.
- Better packaging for spill-prone meals can reduce negative reviews and improve satisfaction, potentially lowering complaint rates by 4–7%.
- Restaurant-specific feedback reports can help partners address service gaps, improving long-term platform quality and vendor performance.

6. Temporal Performance Analysis

6.1 Weekend vs. Weekday Delivery Performance

A paired t-test was conducted to compare delivery times for the same delivery partners across weekday and weekend shifts.



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- A statistically significant difference ($p < 0.05$) confirms weekday delivery times are consistently longer than weekends.
- Peak congestion occurs during lunch and dinner windows, where higher traffic and order density slow fulfillment.

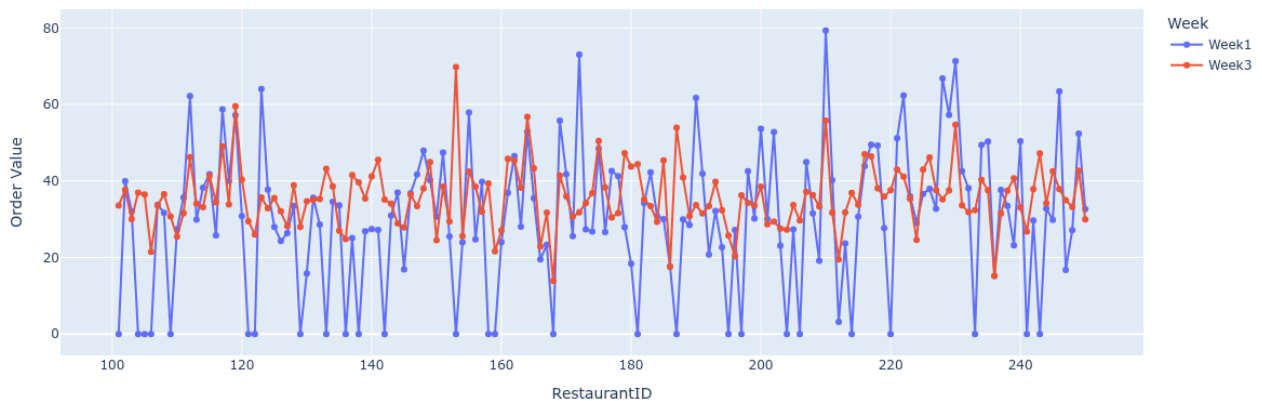
- Moderate variability (SE = 2.45) suggests delays are concentrated around specific high-latency restaurant clusters.

Business Impact Estimation

- Optimizing rider allocation during weekday peak hours can reduce average delivery time by 8–12%.
- Limiting long-distance orders in rush periods may improve on-time delivery rate by 5–7%.
- Early-order incentives can shift demand from peak hours, potentially reducing operational strain by 4–6%.
- Predictive staffing models may improve delivery efficiency and support higher customer satisfaction and repeat orders.

6.2 Before and After Promotional Campaign

A paired t-test comparing Week 1 (pre-campaign) and Week 3 (post-campaign) order values to evaluate promotional effectiveness.



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- Campaign results show a statistically significant increase ($p < 0.05$) in customer spending, confirming the promotion directly improved average order value.
- Week 3 outperformed Week 1 in mean basket size, indicating the campaign effectively influenced purchase behavior.

- The promotion proved effective as a short-term revenue lever across participating restaurant partners.

Business Impact Estimation

- Repeating this campaign model can potentially increase average revenue per order by 8–12% during active periods.
- Extending similar campaigns to lower-rated restaurants may improve both revenue and customer perception, supporting stronger retention.
- Targeting low-frequency customers with recurring offers can increase repeat purchase rates by 5–7%.
- Rotating promotional structures monthly can sustain engagement while reducing customer fatigue and declining campaign response.

7. Strategic Recommendations

Based on your analysis, here's a clear stakeholder-focused recommendations section you can add to the report or presentation.

Recommendation	Priority	Timeline	Expected Impact
Optimize delivery routing in North and West zones	High	Short-term	Reduce average delivery time and improve customer ratings
Increase peak-hour rider allocation during lunch and dinner	High	Short-term	Lower delayed deliveries and improve on-time performance
Redesign VIP membership benefits (priority dispatch, exclusive offers)	High	Short-term	Improve VIP retention and strengthen loyalty
Replicate successful promotional campaign monthly	Medium	Short-term	Increase average order value and platform revenue
Launch targeted offers for regular customers to upgrade to VIP	Medium	Medium-term	Increase customer lifetime value

Focus quality audits on low-rated restaurant partners	High	Medium-term	Improve restaurant ratings and reduce complaints
Introduce cuisine-specific upselling bundles (American, Indian)	Medium	Medium-term	Raise average basket size
Develop predictive staffing by weekday/time-zone demand	High	Long-term	Improve delivery efficiency and operational cost control
Create restaurant performance scorecards for partners	Medium	Long-term	Better vendor management and service consistency
Shift discount strategy toward loyalty bundles instead of blanket discounts	Medium	Long-term	Improve profitability while maintaining repeat orders

8. Expected Outcomes

If implemented, FoodExpress may achieve:

- Faster average delivery times
- Higher customer satisfaction ratings
- Improved repeat order rate
- Stronger VIP retention
- Increased average order value
- Better restaurant performance consistency
- Higher monthly platform revenue

9. Risk & Limitations

Key Limitations

- Analysis uses only **one month of data**, so long-term trends may not be fully reflected.
- External factors like traffic, weather, and public events were not included.
- Customer ratings are subjective and may not always reflect service quality accurately.
- Statistical tests show relationships, not guaranteed cause-and-effect.

Risks

- Seasonal demand patterns may change results.
- VIP-heavy orders may bias customer insights.
- Promotional effects may be temporary.
- Zone-level analysis may miss specific local bottlenecks.

Improvement

- Use 6+ months data
- Add traffic/weather variables
- Include restaurant preparation metrics
- Apply predictive forecasting models

9. Conclusions

The analysis of FoodExpress shows that operational efficiency, particularly delivery speed, has the strongest impact on customer satisfaction and repeat behavior. VIP customers contribute the largest share of orders, but similar satisfaction levels to regular users indicate the need for stronger premium benefits. Promotional campaigns were effective in increasing average order value, proving useful for short-term revenue growth. Geographic differences in delivery performance highlight the need for better staffing and routing in underperforming zones. Overall, improving delivery reliability, enhancing customer loyalty programs, and expanding targeted promotions can significantly strengthen platform performance and long-term customer retention.